# Shruti Sonavane

## AI/ML Engineer

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### PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience in banking and retail, taking machine learning, NLP, and Generative AI systems from prototype to production. Hands-on with RAG pipelines, LLM fine-tuning, vector search, time-series forecasting, and ML infrastructure on AWS and GCP. Recent work includes a compliance-focused RAG system at Citigroup, forecasting models that improved prediction accuracy by 20%, and monitoring practices that cut inference compute costs by 23%.

### EDUCATION

Master of Science in Computer Science, **Indiana University**, **Indianapolis, IN, USA** **August 2024- May 2026**

### PROFESSIONAL EXPERIENCE

**AI/ML Engineer**
**Citigroup, USA** **February 2026 – August 2026**

- Architected and shipped a secure RAG-powered conversational assistant using Llama 3 and LangChain that lets internal audit teams query and summarize financial documents in natural language, reducing manual review time by 65% and improving retrieval accuracy by 28%.
- Built AI copilots and LangGraph-based multi-agent workflows with tool calling and multi-turn context, giving business users governed conversational access to enterprise knowledge, analytics, and decision-support systems.
- Designed guardrails, PII redaction, graceful fallback and error handling, and structured conversation logging so sensitive compliance conversations stay safe, traceable, and debuggable in production.
- Owned conversational quality through LLM evaluation pipelines with prompt versioning and side-by-side testing of prompt and retrieval variants; tracked performance and drift in Grafana and Evidently AI, and retiring degraded models early cut monthly inference compute by 23%.
- Fine-tuned BERT and RoBERTa with LoRA & QLoRA to analyze customer sentiment across multi-channel support conversations, turning customer insights into proactive fixes that improved CSAT by 15%.
- Partnered with Product, Engineering, Analytics, and audit stakeholders to turn strategic ideas into shipped AI features, bringing findings from recent AI research and conferences into team practices.

**Data Scientist**
**Sigma Tech Solutions, India** **August 2021 – July 2024**

- Developed a sales prediction system using a SARIMA forecasting model and deployed it via FASTAPI, improving retail sales forecasting accuracy by 20%.
- Built and deployed a customer churn prediction engine using Random Forest and Scikit-learn, identifying at-risk users with 89% precision enabled the marketing team to launch targeted retention campaigns that recovered 12% of at-risk revenue.
- Scaled NLP annotation pipelines like classification, NER, Sentiment across 2M+ docs/month using Hugging Face Transformers, implemented active learning loop to reduce labelling bottleneck reducing 60% manual labelling effort & pipeline latency under 200ms per document.
- Parallelized nightly batch ETL replacing a sequential 3-hour Airflow DAG using PySpark dynamic partitioning + Airflow task-level parallelism from 3hrs runtime to 45min (-75%) at 10M+ records/day at <1% error rate.
- Built production model monitoring for inference latency, drift & SLA compliance using custom dashboards, auto alerting thresholds, automated rollback triggers which reduced 30% avg incident response time and drift caught within 1 hour of degradation onset.
- Published Tableau and Power BI dashboards on PostgreSQL covering churn, sales forecasts, and model performance for retention and sales teams.

### PROJECTS

**Vendor Mind Agentic RAG Platform**

- Built a conversational, agentic assistant for SaaS spend optimization using Python, SQL, FAISS, and Llama that answers natural-language questions by combining contract evidence with usage analytics.
- Designed conversation flows and tool-calling workflows that extract contract terms, detect data discrepancies, validate renewal constraints, and return clear, actionable recommendations.
- Engineered hybrid retrieval with reranking, clause-aware chunking, verifiable citations, and abstention when evidence is insufficient, keeping agent responses grounded and trustworthy.
- Created an evaluation harness measuring retrieval recall, answer accuracy, citation support, abstention, and latency, and used it to compare prompt and retrieval variants and drive iterative quality gains.

**Credit Card Fraud Detection using Autoencoders in Keras** | *Python, Keras, TensorFlow, Scikit-learn, Pandas, NumPy*

- Built an unsupervised fraud detection model in Keras: an autoencoder trained only on normal transactions, flagging fraud through reconstruction-error thresholds on a 284K-transaction dataset with a 0.17% fraud rate.
- Designed the 4-layer encoder-decoder architecture (14→7→7→29) with L1 regularization and tuned the detection threshold using ROC, precision-recall, and confusion matrix analysis under heavy class imbalance.

**Interactive Astronomical Visualization Platform** | *Python, Streamlit, Plotly, Astropy, Pandas, NumPy*

- Built an ETL pipeline over 600K+ records from ESA Gaia DR3, filtering to the 2,000 brightest stars for fast visualization and analysis.
- Developed a Streamlit app with Plotly and Astropy for real-time sky mapping by geographic location, rendering in 2–3 seconds with PNG, GIF, and MP4 export.

### TECHNICAL SKILLS

**Programming Languages:** Python, TypeScript, SQL
**Conversational AI:** LLM-Powered Agents & Copilots, Conversation Flow Design, Prompt Engineering, Tool Calling, Multi-Turn Context, Guardrails, Fallback & Error Handling, AI Agents, Multi-Agent Systems, Model Context Protocol (MCP)
**Evaluation & Experimentation:** LLM Evaluation Pipelines, Prompt Versioning, Variant / A/B Comparison, Retrieval & Answer-Quality Metrics, Model Monitoring & Drift Detection (Grafana, Evidently AI)
**Frameworks & SDKs:** LangChain, LangGraph, LlamaIndex, CrewAI, OpenAI Agents SDK
**LLM Providers:** OpenAI (GPT-4o/GPT-5), Anthropic (Claude Sonnet/Opus), Gemini, Llama 4
**RAG & Search:** Qdrant, Pinecone, pgvector, Weaviate, FAISS, Hybrid Search, Cohere Rerank, Embeddings
**Fine-Tuning & NLP:** LoRA, QLoRA, Hugging Face PEFT, Unsloth, Axolotl, BERT, RoBERTa, NER, SpaCy, NLTK
**Infrastructure & APIs:** FastAPI, Docker, Kubernetes, AWS EC2/ECS/Lambda/SageMaker, GCP (Dataflow, BigQuery, Vertex AI), GitHub Actions, vLLM, REST APIs, WebSockets / SSE
**ML, Data & Analytics:** PyTorch, TensorFlow, Keras, XGBoost, LightGBM, ARIMA, Scikit-learn, Pandas, NumPy, PySpark, Tableau, Power BI, PostgreSQL, SQL Server